



**MANIPAL INSTITUTE OF TECHNOLOGY**  
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## Parikshit Task -2

# Types of Educational Satellite

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# Introduction

- A **satellite** or an **artificial satellite** is an object, typically a spacecraft, placed into orbit around a celestial body.
- They have a variety of uses, including communication, research, GPS, broadcasting, early warning, earth observation
- The first artificial satellite was Sputnik1.

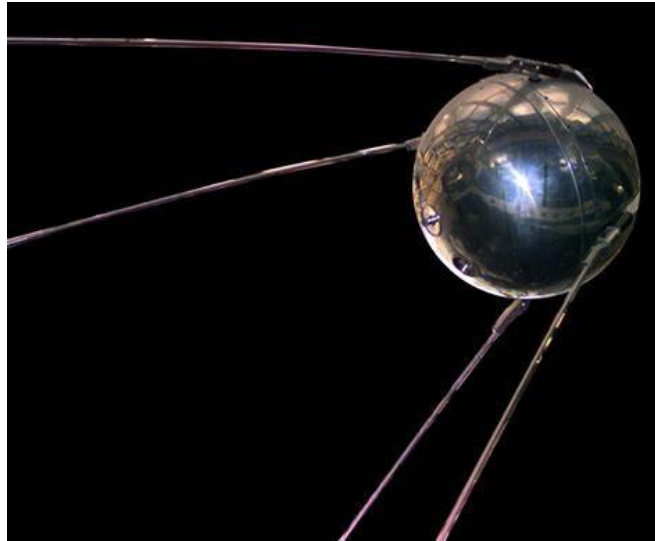
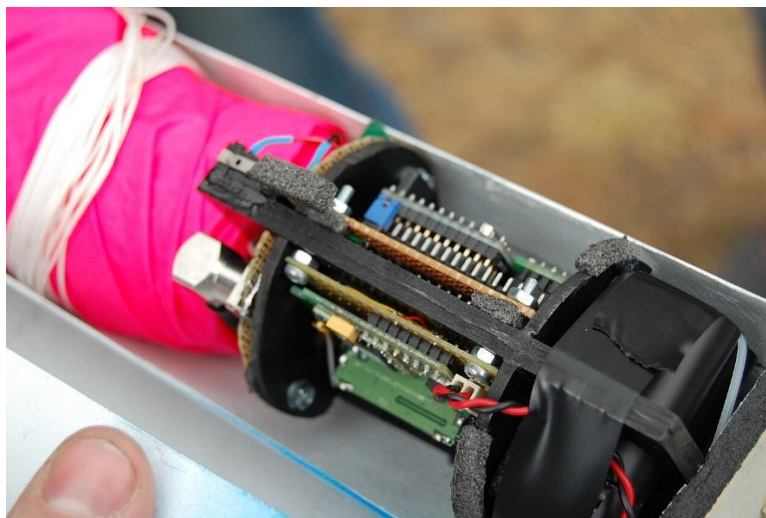


Image 1:Sputnik1first artificial satellite

- ISRO's Student Satellite programme is envisaged to encourage various Universities and Institutions for the development of Nano/Pico Satellites.
- India's first artificial satellite is 'Aryabhata'
- The first educational satellite launched by India is EDUSAT (GSAT-3), which was launched on September 20, 2004.
- It was launched on September 20, 2004

# CanSat

- CanSat is a free educational project run by ESA, the national European Space Education Resource Offices ([ESERO](#)), and various partners across different countries.
- A CanSat is a simulation of a real satellite, integrated within the volume and shape of a soft drink can.
- The challenge for the students is to fit all the major subsystems found in a satellite, such as power, sensors, and a communication system, into this minimal volume.
- The CanSat is then launched by a rocket up to an altitude of approximately one kilometre
- Parachutes are the most common descent device



Internal Image of a Cansat

Source: [CanSat - Wikipedia](#)

## Main elements

Some elements are shared by every CanSat:

### Power supply

- The battery and/or solar panels supply power for operation of all systems. The systems must be able to be powered for four hours continuously and the battery should be easy to access

### **Microprocessor**

- The microprocessor is the heart of the satellite, as it is responsible for receiving signals from external sensors

There are several other components like barometer, thermometer, accelerometer ,gps ,camera to support the mission.

Cansat in Education:

- 1.Low cost of implementation
- 2.Short time for preparation
- 3.Simple design when compared to real satellites
- 4.Wide range of attractive themes
5. These systems typically measure less than 0.15 meters in diameter and 0.4 meters in height and weigh less than 1 kg.

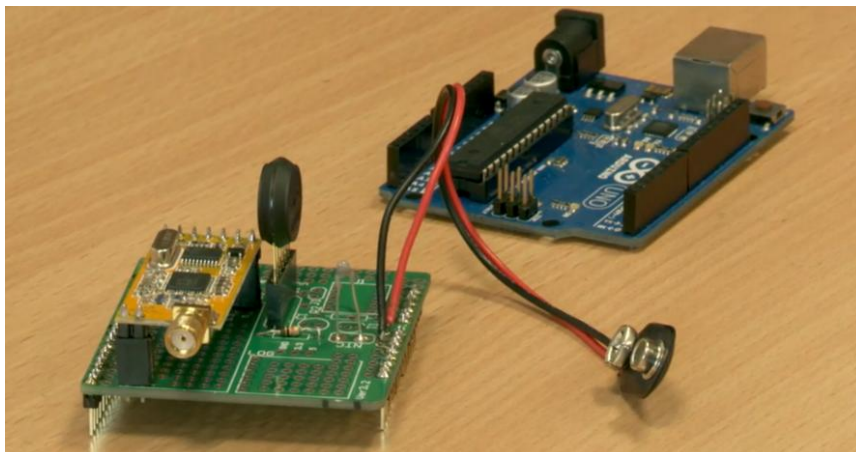


Image of a real Cansat

Source: [CanSat Training Video Part 2- Physical Structure](#)

For a general operation, duration of cansat is 15 minutes

# CubeSat

- CubeSats have a mass of no more than 2 kg
- Scientific experiments with unproven underlying theory may also find themselves aboard CubeSats, because their low cost can justify higher risks
- The standard CubeSat size uses a “one unit” or “1U” measuring 10x10x10 cms and is extendable to larger sizes; 1.5, 2, 3, 6, and even 12U.

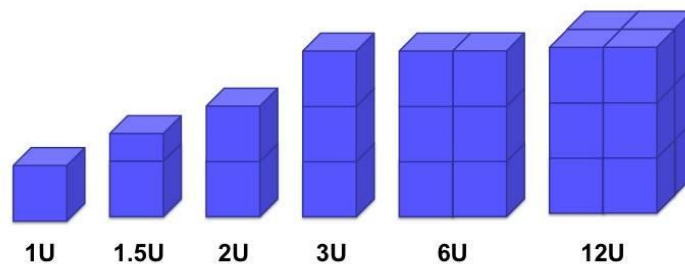


Image: Demonstrating size of cube sat

- NASA Ames launched its first CubeSat, GeneSat, in December 2006.
- Basic 1U CubeSat can cost about \$50,000 to construct. This makes CubeSats a viable option for some schools, universities, and small businesses.
- CubeSat projects usually fall into one of two categories: either you have a specific goal to achieve in orbit (e.g. a business case), or you're part of an academic program where the learning process is just as important as the outcome
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# Flatsat

High Aspect Ratio (HAR) satellites, also known as a FlatSat, represent a forward-thinking approach to satellite design. Their key characteristic is a length more than four times their thickness. This innovative design delivers:

- Enhanced performance due to larger surface areas
- Improved manufacturing efficiency
- Superior drag management in orbit

**Educational Impact:** Makes satellite engineering accessible to students without needing an actual launch.

**Engineering Value:** Provides a safe environment to test CubeSat subsystems before integration.

**Cybersecurity Training:** Prepares future engineers to secure space systems against real-world vulnerabilities.



Image: Flatsat for cubesat testing

# BalloonSat

- Functional Model of a Satellite
- Carried by Weather Balloon into Near Space (> 60,000 Feet)
- Carries Experiments (active and/or passive)
- Collects and Returns Data
- Designed for Launch and Descent Conditions
- Exposed to the Near Space Environment.
- **Altitude:** Typically reach **60,000–100,000 feet** (near-space region).
- **Data Collection:** Records environmental data (temperature, pressure, radiation) and imagery, then transmits or stores it for recovery.
- **Design:** Built to withstand **low pressure, extreme cold, and rapid descent** conditions.
- **Payload:** Can include cameras, sensors (visible, infrared, thermal), GPS, and communication devices.



Image: Depicting launch of balloonsat

# Conclusion

- **FlatSat** provides a **ground-based testbed**. It spreads out satellite subsystems on a flat board, making them accessible for **hands-on testing, debugging, and cybersecurity training**. Students and engineers can practice **commanding, telemetry, power management, and communication systems** without needing a launch.
- **BalloonSat** gives students a **real near-space mission experience**. It is launched by a weather balloon to altitudes above 60,000 feet, exposing payloads to extreme conditions. Students learn about **data collection, payload design, and recovery operations**, simulating a mini space mission.
- CanSat is a **training satellite prototype** designed to fit inside a soda can, giving students a low-cost, hands-on way to simulate a real satellite mission. It is typically launched to modest altitudes by rockets or drones, where it performs descent experiments, collects basic telemetry, and demonstrates mission operations on a small scale. Its simplicity and affordability make it ideal for beginners, offering a practical introduction to aerospace engineering, though it is limited in payload capacity and cannot reach orbit.
- CubeSat is a **modular, standardized mini satellite** capable of conducting real orbital missions in Low Earth Orbit. Built in units of 10×10×10 cm, CubeSats can be scaled up to larger configurations and carry advanced payloads for Earth observation, communication, and scientific research. Unlike CanSats, CubeSats are launched into space by rockets or the ISS, operating for months or years, and bridging the gap between student projects and professional aerospace missions. While more complex and costly, CubeSats provide genuine spaceflight experience and have become a cornerstone of modern small-satellite innovation.

# Summary Table

| Type       | Altitude     | Cost       | Educational Value | Time Duration           | Application                                                                                                     |
|------------|--------------|------------|-------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------|
| CanSat     | 1 kilo meter | Low        | Beginner          | 10–20 minutes           | Teaching basic satellite design, sensors, telemetry, and communication systems                                  |
| Cubesat    | Orbit        | Low-Medium | Real missions     | a few months to 2 years | Real space missions such as Earth observation, scientific research, communication, and technology demonstration |
| Balloonsat | 60k-100k ft  | Low-Medium | Data collection   | Few hours               | Ground testing, debugging, and validation of satellite subsystems before launch                                 |
| Flatsat    | No launch    | Ground     | Testing           | No launch               | Near-space experiments, atmospheric research, environmental data collection, and student training               |

## Reference

1. [FlatSat - High-Aspect Ratio Satellites | Reflex Aerospace](#)
2. [CubeSat - Wikipedia](#)
3. [ESA - CanSat 2025-2026: Challenge your students to build a can-sized satellite](#)
4. [CanSat - Wikipedia](#)